DSN AI Bootcamp 2026 Project Participation – Data Analytics Track

Answer & Submission Sheet

Participant Name	**Nathan Aguele**
Dashboard tool used	**Power BI**

Part A - Data Cleaning Review

Identify at least three meaningful data-quality issues you found. For each issue, explain what you found, what you did, and why your action was appropriate.

Data-quality issue	What you found	What you did	Why this action was appropriate
Category & Casing Inconsistency with trailing whitespace	*I found that same values were differentiated between, due to category and casing errors like Accessory/Accessories and Lagos/LAGOS and extra whitespace*	*I formatted these values to one standard form using text transformations and uniform naming conventions, singular or plural, not both.*	*The action was appropriate because it ensured consistency and accuracy across state and category columns*
Revenue & Profit Reported Calculation errors	*The revenue reported in the original dataset failed to apply discount rates properly, which in turn lead to inflated values for Revenue reported and Profit; 47,500,870 and 16,617,270 respectively*	*I recalculated revenue using the formula "Unit_Price * Quantity * (1-Discount_Pct)", I then used this recalculated values to recompute the profit*	*It ensured transparency and accuracy as it gives management the true values of revenue and profit: 46,679,725 AND 14,590,425*

Null/missing values	*The "Cost_Per_Unit_NGN" column had missing/null values which in turn affected subsequent calculations as any queries made would treat these values as 0 giving phantom profits*	*I recomputed these values using a conditional column to replace null/missing values with the appropriate values*	*It ensured accuracy of my subsequent calculations, prevented analysis of phantom profits and revenue and stops data errors*

Wrong data types and formats applied to columns	*I found that a lot of the columns had incorrect data types, like most of the columns that should have been currency were instead set to numbers and columns that should have been percentage were set to numeric decimals.*	*I formatted each individual column to a correct and coherent data type, columns that directly documented financial transactions were set to currency, columns that represent names, set to text and columns that represent date, were set to date only as the exact time was "00:00".*	*I formatted and corrected the data in this way because it ensures accurate calculations, improves the readability of the data and prevents data errors.*

Which data-quality issue had the greatest impact on your analysis?

The data quality issue that had the most impact in my analysis were the data transformations, the inconsistency and incorrectness of certain values and columns led to severe breaks in data quality and overall analysis

Explain how correcting that issue changed or affected the analysis.

Correcting these values and columns gave my data set proper structure and validation, it ensured that any data displayed from my dashboard would not have any misleading and incorrect facts like "Abuja was the most profitable state".

Part B - Dashboard Audit

For each management claim, state your verdict, provide the corrected result where applicable, and explain the evidence supporting your conclusion.

1. Abuja is the most profitable market.

Verdict

Type: Correct / Incorrect / Misleading / Cannot be determined

Corrected answer / result

Incorrect

Explanation / evidence

Abuja was the least profitable market as it realized only ₦1.7 million in profits and ₦ 1.46 million after returns, these values were realized after standardization of columns and their values. See the Audited Dashboard for more details

2. Electronics is the strongest-performing product category and should receive more budget.

Verdict

Type: Correct / Incorrect / Misleading / Cannot be determined

Corrected answer / result

Correct

Explanation / evidence

Electronics performed significantly better than other categories, netting ₦ 29 million in revenue and ₦ 25.5 million after returns

3. Higher discounting is improving profitability.

Verdict

Type: Correct / Incorrect / Misleading / Cannot be determined

Corrected answer / result

Misleading and Incorrect

Explanation / evidence

Profits degraded even with higher discounts, there was no real uptick in units bought with respect to discount given, units bought by customers who were given no discount were the largest among all other customers

4. The Online channel has the lowest return rate.

Verdict

Type: Correct / Incorrect / Misleading / Cannot be determined

Corrected answer / result

Incorrect

Explanation / evidence

The online channel had a return of 9.70% which was the second highest return rate. The channel with the lowest return rate was "In-Store" with a return rate of 6.44%.

5. August is the strongest sales month.

Verdict

Type: Correct / Incorrect / Misleading / Cannot be determined

Corrected answer / result

Incorrect

Explanation / evidence

The strongest sales month, going solely by revenue was May with ₦7 million, after returns it was January with ₦6.3 million. January also had the highest amount of units sold among any other month with 174 units sold.

6. Lagos is underperforming compared with other markets.

Verdict

Type: Correct / Incorrect / Misleading / Cannot be determined

Corrected answer / result

Incorrect

Explanation / evidence

Lagos is considerably the highest performing state compared to other markets, with approximately ₦4.6 million in profits, ₦4.2 million after returns and had the highest units sold compared to any other month,

7. The reported Revenue field can be used directly for decision-making.

Verdict

Type: Correct / Incorrect / Misleading / Cannot be determined

Corrected answer / result

Incorrect

Explanation / evidence

The reported revenue of the original dashboard did not factor for discounts or returns, leading to a phantom value of ₦47,500,870. After cleaning and validation, revenue was ₦46,679,725 and ₦42,031,350 after returns

8. The dashboard is sufficiently reliable for management to act on without further data cleaning.

Verdict

Type: Correct / Incorrect / Misleading / Cannot be determined

Corrected answer / result

Incorrect

Explanation / evidence

The dashboard had a lot of error and flaws, acting on it without further data cleaning would have led to erroneous decisions by management.

Part C - Corrected Dashboard

Insert a clear snapshot of your corrected dashboard below. The dashboard may be created in Power BI or Excel.

AUDITED MANAGEMENT DASHBOARD

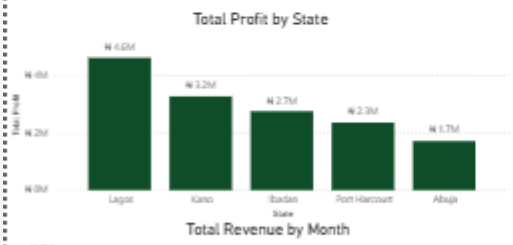
₦ 46.68M	₦ 14.59M	8.57%	7.99%
Total Revenue	Total Profit	Return Rate	Average Discount

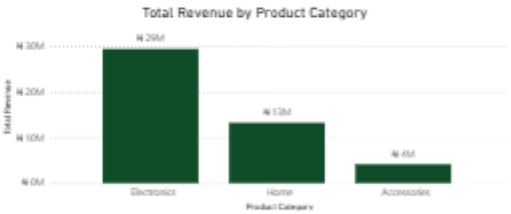
Product Category	Total Revenue
Electronics	₦ 29,346,500.00
Home	₦ 13,206,300.00
Accessories	₦ 4,126,925.00
Total	**₦ 46,679,725.00**

State	Total Profit
Lagos	₦ 4,579,900.00
Kano	₦ 3,243,600.00
Ibadan	₦ 2,730,525.00
Port Harcourt	₦ 2,338,500.00
Abuja	₦ 1,698,100.00
Total	**₦ 14,590,425.00**

Channel	Return Rate
Whatsapp	11.90%
Online	9.70%
In-Store	6.44%
Total	**8.57%**

Month	Total Revenue
January	₦ 6,722,250.00
February	₦ 5,364,300.00
March	₦ 6,298,725.00
April	₦ 6,343,000.00
May	₦ 7,008,500.00
June	₦ 5,397,250.00
July	₦ 6,378,275.00
August	₦ 3,167,425.00
Total	**₦ 46,679,725.00**









AUDITED MANAGEMENT DASHBOARD

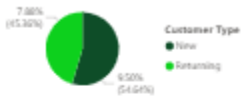
₦ 42.03M
Revenue after Returns

₦ 13.15M
Profit after Returns

Revenue Vs Return



Return Rate by Customer Type



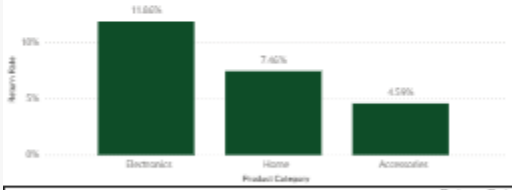
Product Category	Revenue after Returns
Electronics	₦ 25,555,750.00
Home	₦ 12,450,600.00
Accessories	₦ 4,025,000.00
Total	**₦ 42,031,350.00**

State	Profit after Returns
Lagos	₦ 4,280,475.00
Kano	₦ 2,666,950.00
Ibadan	₦ 2,497,925.00
Port Harcourt	₦ 2,245,900.00
Abuja	₦ 1,459,900.00
Total	**₦ 13,151,150.00**

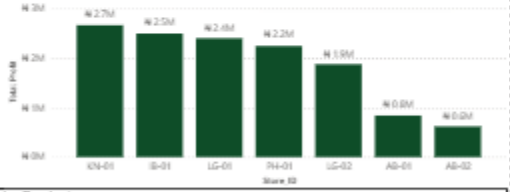
Channel	Total Return
In-Store	₦ 2,114,375.00
Whatsapp	₦ 1,346,000.00
Online	₦ 1,188,000.00
Total	**₦ 4,648,375.00**

Month	Revenue after Returns
January	₦ 6,329,800.00
February	₦ 4,552,900.00
March	₦ 6,061,825.00
April	₦ 6,209,275.00
May	₦ 6,145,900.00
June	₦ 5,080,250.00
July	₦ 5,436,525.00
August	₦ 2,212,875.00
Total	**₦ 42,031,350.00**

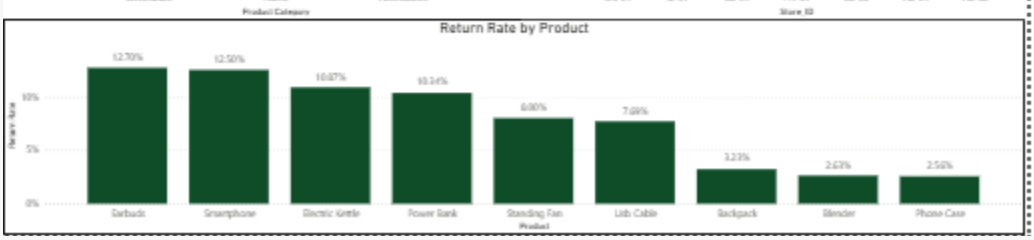
Return Rate by Product Category



Total Profit by Store



Return Rate by Product



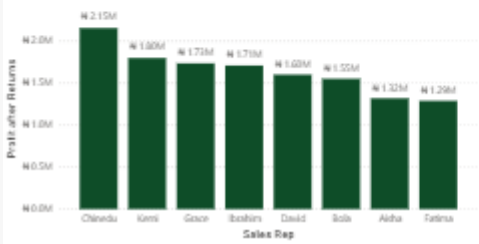
Return Rate by Sales Rep



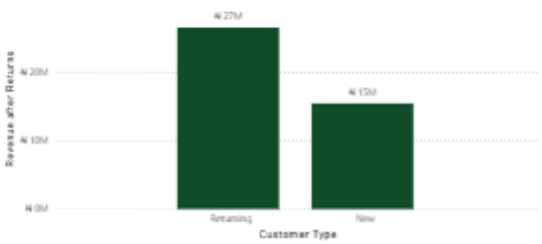
Revenue after Returns by Sales Rep

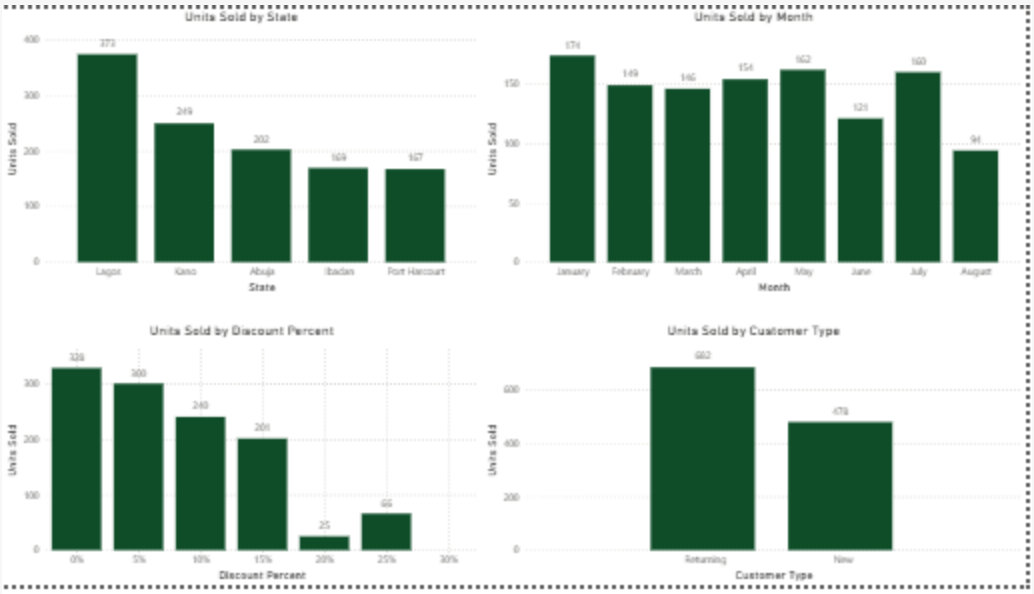


Profit after Returns by Sales Rep



Revenue after Returns by Customer Type




Units Sold by State
Units Sold
400
300
200
100
0
271
249
202
169
167
Lagos
Kano
Abuja
Ibadan
Port Harcourt
State
Units Sold by Month
Units Sold
150
100
50
0
174
149
146
154
162
121
160
94
January
February
March
April
May
June
July
August
Month
Units Sold by Discount Percent
Units Sold
300
200
100
0
329
300
240
201
25
66
0%
5%
10%
15%
20%
25%
30%
Discount Percent
Units Sold by Customer Type
Units Sold
600
400
200
0
682
478
Returning
New
Customer Type

Part D – Final Analysis Questions

1. Which conclusion in the original dashboard was the most misleading, and why?

Answer

From the management summary, the most misleading conclusions were Abuja being the most profitable market and higher discounting improving profitability.

Explanation / evidence

Abuja was the least profitable market netting only profits of ₦1.7 million and ₦1.46 million after return. Discounts did not improve profitability as units sold were marginally higher among those who were given no discount.

2. What changed most materially after you cleaned and corrected the data?

Answer

Revenue and Profits before and after Returns

Explanation / evidence

Revenue and profits became significantly lower after accounting for discounts and returns, see figures 1 & 2 for more details

3. If management had acted on the original dashboard, what decision could have created the greatest business risk?

Answer

Investing more in the Abuja market

Explanation / evidence

The Abuja was the least profitable market, investing would have been equivalent to pouring gold coins down the drain.

4. Based on your corrected analysis, what is the most important driver of business performance in this dataset?

Answer

The most important driver of business performance was the financial driver

Explanation / evidence

Business performance relied mainly on the revenue and profit generated, product costs and discounts given with respect to units sold and profit generated.

5. What three actions should management take next based on the evidence from your corrected dashboard?

Answer

1. Invest in the stronger markets, categories and channels
2. Keep Discounts at a minimum and play to seasonal strengths
3. Figure out reason for returns and fix what needs to be fixed

Explanation / evidence

1. *Management should invest in the markets that generate significant revenue, while Lagos brought in the highest revenue, the Kano store and Ibadan store generated more than any single Lagos store, this implies that opening a second location in both states could significantly drive-up revenue. Electronics were the best seller, bringing in significantly more profits than any other category, investing more in it would be the best choice. Investing more in your "In-Store" channel would be a good idea, as they had the least return risk and generated the best profit, weaker channels like "Whatsapp" had the greatest return risk and the least profit overall.*
2. *There is no proven increase in demand/units sold due to higher discounts so its best to keep discounts at a minimum to reduce losses and increase profitability. The months of March to May generated consistently high revenue to you can play those key seasonal markets to your strengths.*
3. *Ask your customer about the reasons for returns, if it was due to personal reasons or by reason of product defects, if it was by reason of defects, management should seek to rectify these defects to reduce returns and losses from returns*

Submission reminder: Complete all sections, insert a readable corrected-dashboard snapshot, ensure your written figures match the dashboard, and keep your Power BI or Excel working file available if verification is requested.